

COURSE TITLE : **CROSS PLATFORM MOBILE APPLICATION DEVELOPMENT**
 COURSE CODE : **6263**
 COURSE CATEGORY : **A**
 PERIODS/WEEK : **5**
 PERIODS/SEMESTER : **75**
 CREDITS : **5**

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	HTML5 & CSS3	18
2	Ionic frameworks	19
3	AngularJS	19
4	Apache Cordova	19

Course General Outcomes:

Sl.	G.O	On completion of this course the student will be able to :
1	1	Develop cross platform mobile application development using HTML5 & CSS3
2	1	Develop cross platform mobile application development using Ionic frameworks
3	1	Develop cross platform mobile application development using AngularJS
4	1	Develop cross platform mobile application development using Apache Cordova

Specific Outcomes:

MODULE – I: HTML5 & CSS3

1.1Develop cross platform mobile application development using HTML5 & CSS3

- 1.1.1 Define Mobile applications
- 1.1.2 Describe cross platform Mobile applications
- 1.1.3 Discuss History of Mobile applications and platforms
- 1.1.4 Discuss Current mobile platforms
- 1.1.5 List Development tools used in mobile application
- 1.1.6 List Languages used in mobile application
- 1.1.7 Define HTML5

- 1.1.8 List New features in HTML5
- 1.1.9 Explain HTML5 based Mobile Applications
- 1.1.10 List HTML5 API features
- 1.1.11 Explain defining new elements, new attribute syntax, semantic elements, input elements in HTML5
- 1.1.12 Explain Canvas, Scalable Vector graphics, media Types in HTML5
- 1.1.13 Discuss HTML5 geo location API
- 1.1.14 Discuss HTML5 local storage
- 1.1.15 Define CSS3
- 1.1.16 List New Features in CSS3
- 1.1.17 Explain CSS3 Selectors
- 1.1.18 Discuss CSS3 box model, CSS3 borders, CSS3 backgrounds
- 1.1.19 Discuss CSS3 new colours, gradients, shadows
- 1.1.20 Describe CSS3 transformation, 2D and 3D transformations, Transitions, Animations
- 1.1.21 Explain CSS3 box sizing, CSS3 Flex boxes

MODULE – II: Ionic frameworks

2.1 Develop cross platform mobile application development using Ionic frameworks

- 2.1.1 Discuss to Mobile user Interface Development
- 2.1.2 List UI development Frameworks
- 2.1.3 Discuss Ionic Framework
- 2.1.4 List Support and Features of Ionic Framework
- 2.1.5 Demonstrate Installing Ionic frameworks, platform supports (iOS and Android)
- 2.1.6 Demonstrate Creating a User Interface project in Ionic
- 2.1.7 Explain Ionic elements, lists, menus, slide boxes, refreshing
- 2.1.8 Discuss Ionic NetBeans
- 2.1.9 Demonstrate Building Ionic UI applications
- 2.1.10 Demonstrate Testing Ionic Applications

MODULE – III: AngularJS

3.1 Develop cross platform mobile application development using AngularJS

- 3.1.1 Define AngularJS
- 3.1.2 Summarize angularJS MVC architecture
- 3.1.3 Explain angularJS controllers, angularJS views, angularJS models
- 3.1.4 Describe angularJS expressions, directives, filters

- 3.1.5 List angularJS http object and functions
- 3.1.6 Describe angularJS event Management
- 3.1.7 Explain angularJS SQL management
- 3.1.8 Discuss angularJS Modules
- 3.1.9 Describe angularJS API

MODULE – IV: Apache Cordova

4.1 Develop cross platform mobile application development using Apache Cordova

- 4.1.1 Define Apache Cordova
- 4.1.2 Discuss history & Features of Cordova
- 4.1.3 List Basic Requirements of Cordova
- 4.1.4 Describe Creating cross-platform applications using HTML5-CSS3-javascript based frameworks in Apache Cordova
- 4.1.5 List Apache Cordova plug-ins for iOS, Android and Windows based Applications
- 4.1.6 List Technologies used along with Apache Cordova
- 4.1.7 Demonstrate how to create a project in Apache Cordova
- 4.1.8 Explain Building Applications in Apache Cordova
- 4.1.9 Distinguish iOS, Android and Windows application Compilation
- 4.1.10 Describe Setting up Application files, Managing Local Storage Options
- 4.1.11 Describe Single Page Applications, Managing UI Templates, view classes, view routing
- 4.1.12 Discuss Managing Location API, Contacts API, Camera API
- 4.1.13 Discuss Testing Applications

CONTENT DETAILS

MODULE – I:

Introduction to Mobile applications – History of Mobile applications and platforms – Discussion on Current mobile platforms - Development tools used in mobile application – Languages used - Introduction to HTML5 – New features in HTML5 - HTML5 based Mobile Applications - HTML5 API features – Defining new Elements – New attribute syntax in HTML5 - Semantic Elements in HTML5 – HTML5 input elements – Canvas – Scalable Vector graphics – HTML5 media Types - HTML5 geo location API - HTML5 local storage - Introduction to CSS3 – New Features in CSS3 - CSS3 Selectors - CSS3 box model – CSS3 borders – CSS3 backgrounds - CSS3 new colours –

gradients – shadows - CSS3 transformation – 2D and 3D transformations – Transitions – Animations - CSS3 box sizing - CSS3 Flex boxes

MODULE – II:

Introduction to Mobile user Interface Development – UI development Frameworks - Introduction to Ionic Framework – Support and Features - Installing Ionic frameworks – platform supports (iOS and android) - Creating a User Interface project in Ionic - Ionic elements – lists – menus – slide boxes – refreshing - Ionic NetBeans - Building Ionic UI applications - Testing Ionic Applications.

MODULE – III:

Introduction to AngularJS – Features in angularJS - angularJS MVC architecture - angularJS controllers - angularJS views - angularJS models - angularJS expressions -directives – filters - angularJS http object and functions - angularJS event Management - angularJS SQL management - angularJS Modules - angularJS API

MODULE – IV:

Introduction to Apache Cordova – history - Features - Basic Requirements - Creating cross-platform applications using HTML5-CSS-javascript based frameworks in Apache Cordova - Apache Cordova plug-ins for iOS, Android and Windows based Applications. - Technologies used along with Apache Cordova - Creating a Project in Apache Cordova - Building Applications in Apache Cordova – iOS, Android and Windows application Compilation - Setting up Application files – Managing Local Storage Options – Single Page Applications - Managing UI Templates – view classes – view routing - Managing Location API – Contacts API – Camera API - Testing Applications

TEXT BOOK(S)

1. Building Android Apps with HTML, CSS, and JavaScript, 2nd Edition, By Jonathan Stark, Brian Jepson, Brian MacDonald, Publisher: O'Reilly Media

REFERENCE

1. Beginning Android Web Apps Development: Develop for Android using HTML5, CSS3, and JavaScript, By Jon Westfall , Rocco Augusto , Grant Allen, Publisher: Apress